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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester M.C.A. (CBCS) Degree Examinations
January 2024

22MCA331 – MOBILE COMPUTING AND APPLICATION DEVELOPMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from each **Unit**

	Unit – I	Marks	BT*	CO*	PO*
1. a)	What is Mobile computing? Explain mobile computing functions with a neat diagram.	8	L*2	1	1
b)	Explain the GPRS network architecture.	8	L2	1	1
c)	Write a short note on Fourth Generation telecommunication technology.	4	L2	1	1
2. a)	Elaborate different types of middleware that are used in mobile computing environment.	7	L2	1	1
b)	Explain the Three tier Architecture for Mobile computing.	3	L2	1	1
c)	Explain the different entities of GSM architecture.	4	L2	1	1
d)	Explain the Sequence of Operation from Speech to Radio Wave in GSM with the help of an example.	6	L2	1	1
	Unit – II				
3. a)	What is Android operating system and why is it a popular mobile operating system for application development?	5	L1	2	1
b)	Explain how to Run the application on an emulator or a device.	5	L2	2	1
c)	Describe the Android application architecture and how it supports the development of diverse applications.	10	L2	2	1
4. a)	Explain the development process of an application on Android studio.	4	L2	2	1
b)	Explain the usage of a ScrollView with a LinearLayout and provide an example.	6	L2	2	1
c)	What is the use of LOG class in Android? Explain.	4	L2	2	1
d)	Explain TextView widget in Android and its attributes in detail.	6	L2	2	1
	Unit – III				
5. a)	What is activity? Explain activity navigation in Android.	5	L2	3	1
b)	Explain the differences between explicit and implicit intents in Android.	5	L2	3	1
c)	Explain the Passing of data from one Activity to another and Getting data back from an Activity.	10	L2	3	1
6. a)	Illustrate the Android activity lifecycle and explain the purpose of key methods like onCreate , onStart , onResume , onPause , and onDestroy .	10	L2	3	1
b)	Explain the 'Activity attributes' and 'Intent flags' related to Activity launch modes in detail.	5	L2	3	1
c)	Explain the configuration changes and activity changes in Android.	5	L2	3	1
	Unit – IV				
7. a)	Explain the usage of the following with an example for each.				
i)	Radio Button				
ii)	Checkbox				
iii)	Spinner				
iv)	Toggle button and switch.	10	L2	4	1
b)	Explain the steps used to create Popup menu in Android.	6	L2	4	1
c)	What you mean by Pickers? Explain Date and Time Pickers.	4	L2	4	1

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| 8. | a) | Explain the different steps which are used to implementing RecyclerView. | 10 | L2 | 4 | 1 |
| | b) | What are the key advantages of using drawables in Android UI design? Explain. | 10 | L2 | 4 | 1 |

Unit – V

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| 9. | a) | What are is Broadcast messages? Explain different types of Broadcasts messages. | 5 | L2 | 5 | 1 |
| | b) | What are content providers? Explain. | 5 | L2 | 5 | 1 |
| | c) | What are Broadcast receivers? Explain the types of Broadcast receivers with example. | 10 | L2 | 5 | 1 |
| 10. | a) | Build a GUI for Student table with 5 records, write queries to add, delete and view the records from the SQLite database. | 10 | L2 | 5 | 1 |
| | b) | Write a short note on Service app component. | 4 | L2 | 5 | 1 |
| | c) | How do you implement a SQLite database? Explain with an example. | 6 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
